

Dr. Ashish K Sharma

Professor of Physics

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Ashish K Sharma is Professor of Physics at the Central University of Jammu, where he leads the Energy Materials and Devices Laboratory. After a doctorate at IIT Delhi in 2013, he spent eight years at the Inter-University Accelerator Centre in New Delhi, first as a DST-INSPIRE Faculty Fellow and then as a SERB Research Scientist. He taught at UPES Dehradun before moving to Jammu in 2023.

His first research thrust is energy conversion. The group works on thermoelectric materials in all their forms — inorganic, organic, composite, flexible and printable — and builds them into devices that make electricity from waste heat. Four funded projects have come out of this line, along with the RSC Advances Outstanding Student Paper Award in 2024 and a Best Ph.D. Thesis Award in 2025.

The second is semiconductor devices built on strategic materials: gallium nitride, gallium oxide and silicon carbide, for electronics that must hold up at high power, at high temperature, at low noise, and in radiation-harsh environments. His group improves them through surface passivation, annealing and controlled processing, and then tests how far that tolerance extends.

Behind both sits the question of how defects form and evolve, which he approaches with experiment and theory together, and which now carries into one- and two-dimensional nanomaterials and heterostructures for space applications.

He is also a builder of instruments. His laboratory has developed low-cost measurement systems for the Seebeck coefficient, resistivity, thermal conductivity, gas sensing and in-situ deep level transient spectroscopy, several of which have few equivalents in the country.

He has published 91 papers, holds three patents, and has an h-index of 27 with close to 1,860 citations. His work has drawn ₹2.05 crore in funding from DST, SERB, ISRO, DRDO and IUAC. In 2019 he was one of fewer than 600 young scientists chosen worldwide for the Lindau Nobel Laureate Meeting.

He has guided seven doctoral scholars and more than twenty master's students, who value him as much for his teaching and his readiness to help as for the physics.

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